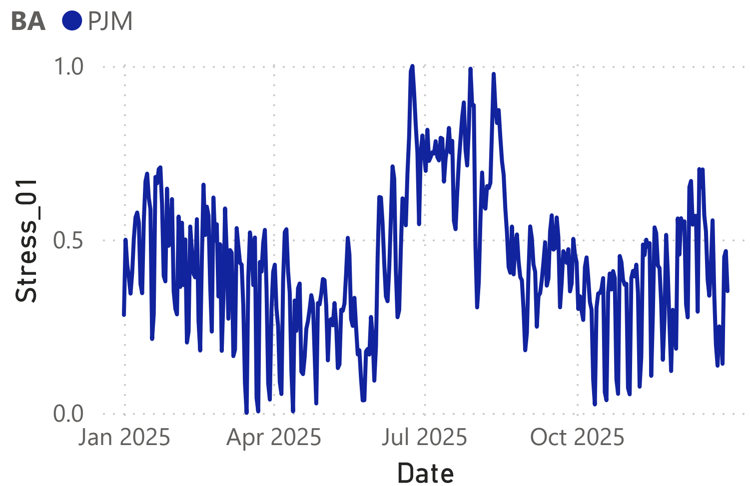
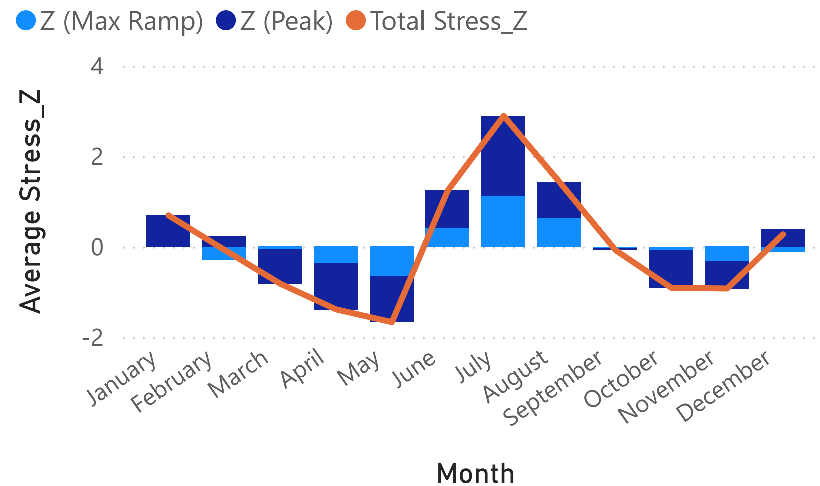
Electric Grid Stress Analysis - Overview

This dashboard visualizes a machine learning framework built on public grid data classifying daily stress regimes across multiple Balancing Authorities (BAs) using z-scores

Daily Stress_01



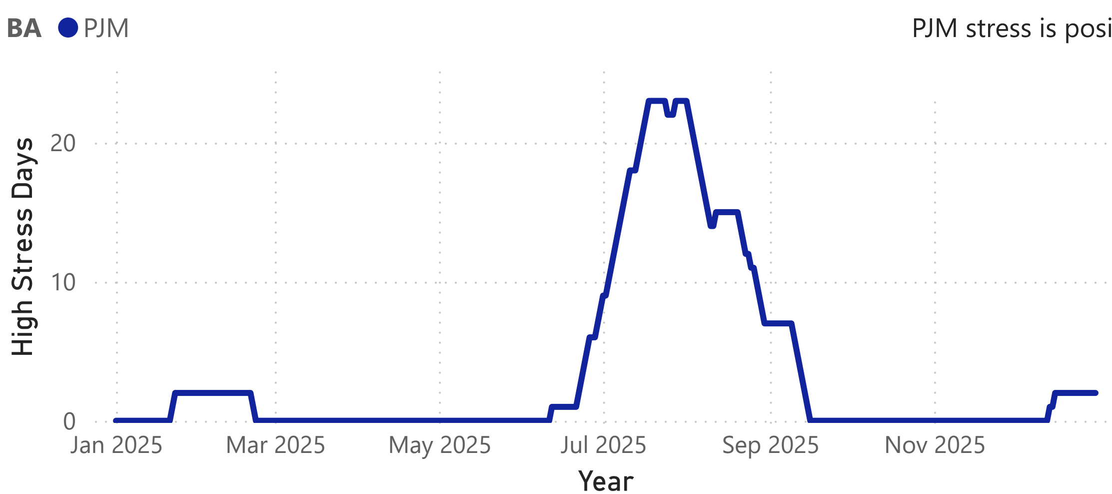
Average Stress by Month



Note: Stress by Month only informative by separated BAs

PJM stress is positive in winter months

Number of High Stress Days in the Previous 30 Days



Both systems peak in summer, but CAISO's stress concentrates into a tight Aug–Sep window while PJM builds earlier and holds longer

<u>Analytic Structure</u>

Peak = Daily max power demand
Max_Ramp: Daily max power ramp between consecutive hours
Stress_Z = Z_Peak + Z_Max_Ramp (mean = 0, value ≈ std dev)
Stress_01 = normalized Stress_Z by BA (min = 0, max = 1)

Balancing Authority (BA)

Select all | CISO | PJM

Most Stressful Days Table

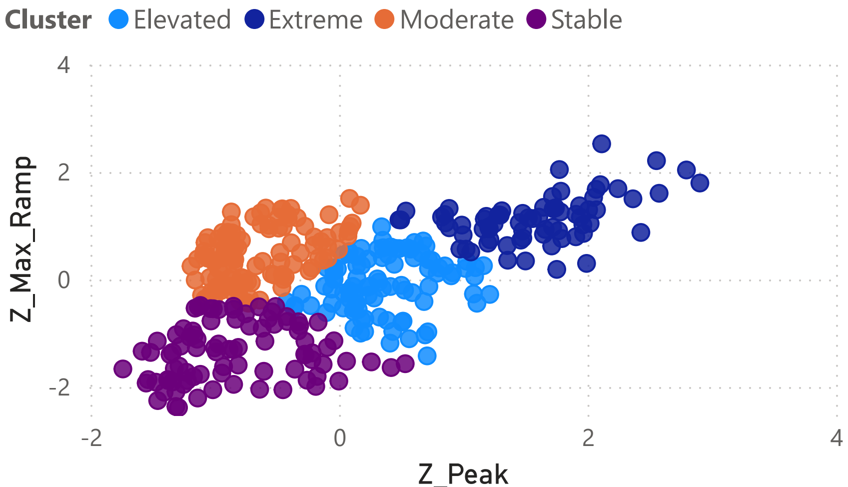
BA	Stress_01	Date
PJM	0.87	August 15
PJM	0.88	August 13
PJM	0.89	July 30

Stress_Z (Max)	Stress_01 (Avg)	Most Stressed Day	High Stress Days
PJM	PJM	PJM	PJM
4.83	0.43	Jun 24	46

CISO has higher stress days but PJM has more high stressed days
- High Stress Days Threshold: Z_01 > 0.7

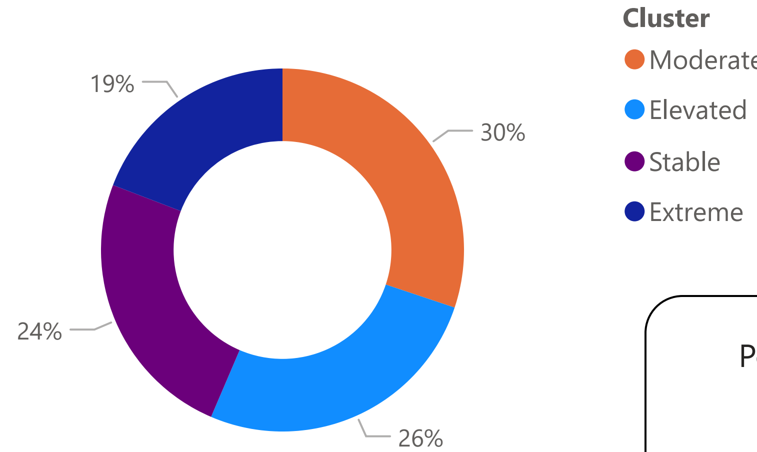
Electric Grid Stress Analysis - ML Clustering

Z-Score Clustering



Both BAs cluster similarly with instability increasing along the diagonal.
CISO reaches higher extremes with more spread out datapoints

Frequency by Cluster



PJM contains more than double the extreme days
CISO contains nearly double the stable days

Balancing Authority

CISO | PJM

Operators can use this calculator to determine how stressed their grid would be given specified conditions

Daily Stress Calculator

Peak Input (MW)	Max Ramp Input (MW)
100000	7000

Moderate
Identified Cluster

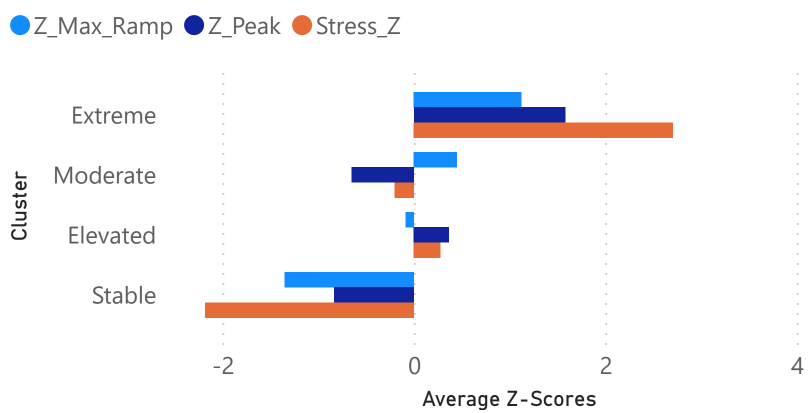
62%
Stress Percentile

0.40
Input Stress Z

-0.20
Cluster Mean Stress Z

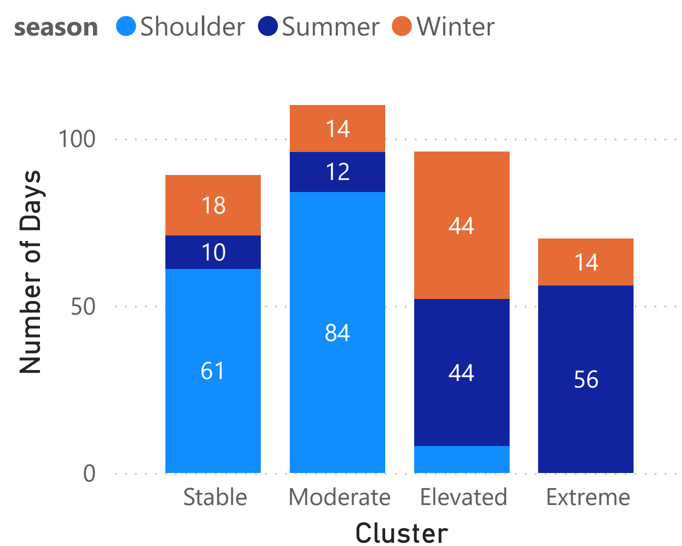
Approximate 2025 Ranges (MW)
CISO Peak: 23,000 - 45,000 || CISO Ramp: 1,000 - 5,000
PJM Peak: 78,000 - 160,000 || PJM Ramp: 2,500 - 9,100

Average Z-Scores by Cluster



CISO's Extreme cluster combines peak and ramp stress nearly equally
- both must be elevated simultaneously. PJM's Extreme cluster is more peak-led (z_peak 1.58 vs z_ramp 1.12).

Number of Days by Cluster and Season



PJM has more extreme days (including winter days)

k	CISO	PJM	Total
2	0.51	0.42	**0.93**
3	0.40	0.41	**0.81**
4	0.40	0.41	**0.81**
5	0.38	0.39	**0.76**
6	0.37	0.41	**0.79**
Total	**2.06**	**2.04**	**4.09**

Model Validation - Silhouette Scores

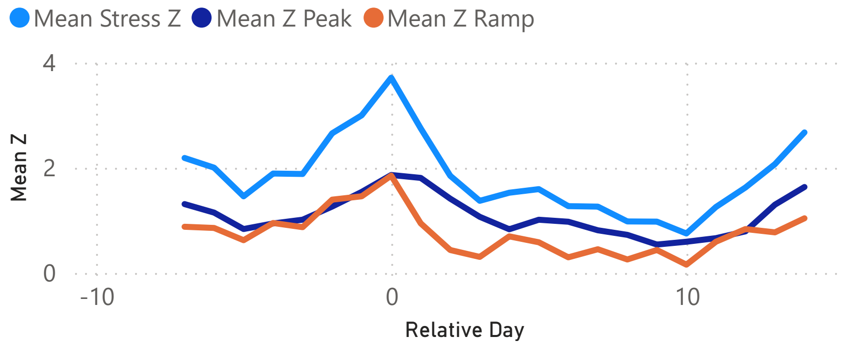
k=4 best for operational interpretability
- four regimes map well to Stable → Extreme
- k=3 collapses the moderate range;

k=6 adds granularity without adding insight

Electric Grid Stress Analysis - Event Drilldown

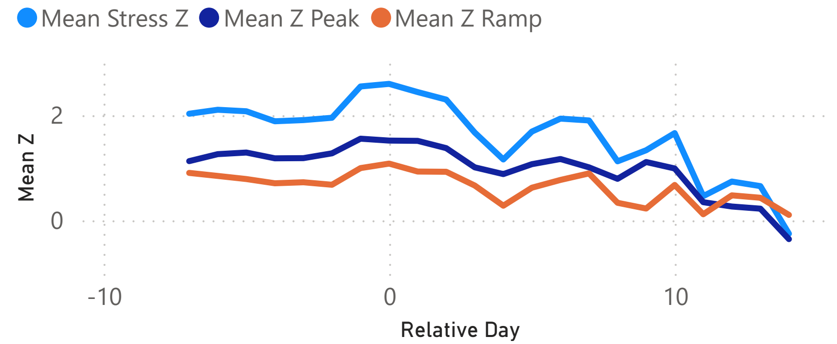
Composite events - An average of the extreme stress events. How stress builds and resolves around extreme events (day 0 = stress peak)

Composite Extreme Event (CAISO Summer)



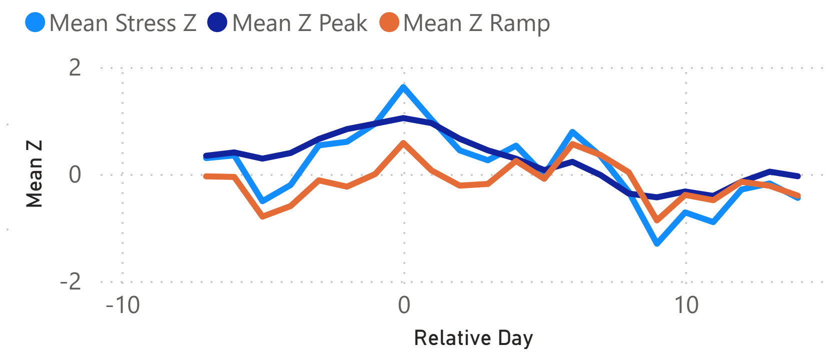
Ramp-amplified— solar drop-off drives the evening stress spike
- sharp ramp drives stress on extreme day when solar drop-off occurs

Composite Extreme Event (PJM Summer)



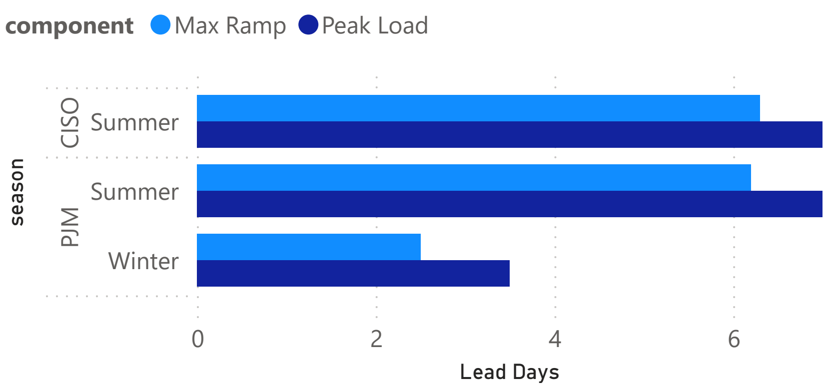
Load-dominated - slower build-up and drop off due to heat dome effect

Composite Extreme Event (PJM Winter)



Load-dominated — quick cold snap raises demand
- ramp signal is weak or negative

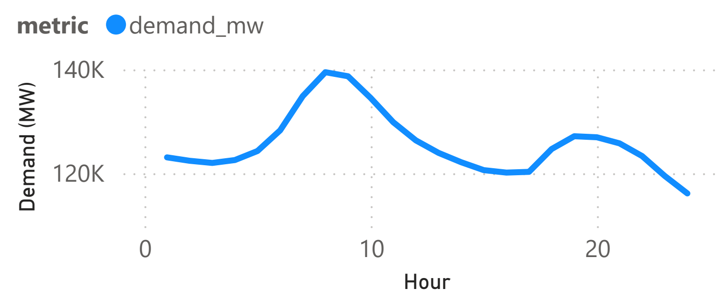
Lead Days to Extreme Event



Above: Summer events give much more warning than winter PJM events

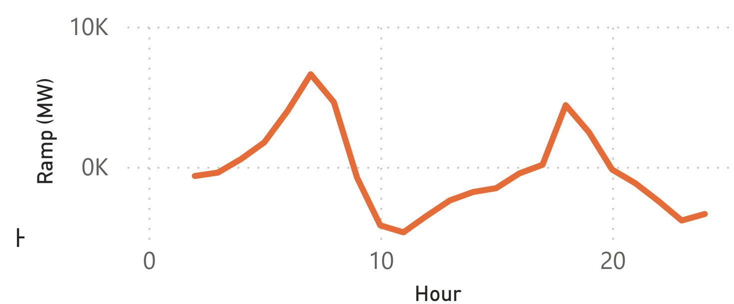
Right - Hourly differences by BA and season demonstrated
- i.e. CISO summer duck curve and PJM winter heating ramps
- CISO solar penetration is ~25% but PJM solar penetration is only ~5%

Hourly Demand on Extreme Day



CISO | PJM | Summer | Winter

Hourly Ramp on Extreme Day



Summary notes:

- CAISO stress is acute and ramp-amplified — few days, very high peaks. PJM stress is broader and load-led — more days, sustained demand. Different mechanisms require different responses.
- Compound event — heat wave elevates load for days; solar drop-off then triggers the ramp spike that pushes to Extreme"
- The additive stress formula can produce near-zero stress scores for high peak / low ramp days.
- In practice, such days are rare (r=0.65 between components), but the formula is not designed to flag isolated capacity-only stress.
- All data is based on 2025, but analysis can be applied to other years.